1. What is Static Analysis?

Answer:
Static analysis is the process of analyzing source code without executing it, mainly to detect bugs, enforce coding standards, and improve code quality early.

---

2. What is the difference between Static Checking and Dynamic Checking?

Answer:
Static checking finds errors before execution (compile time), while dynamic checking finds errors during execution (runtime).

---

3. Explain Static Typing in Java.

Answer:
Static typing means that variable types are known at compile time, allowing the compiler to detect type-related errors early.

---

4. Give examples of errors caught by Static Checking.

Answer:
Syntax errors, wrong method names, wrong number of arguments, type mismatches, and incorrect return types.

---

5. Give examples of errors caught by Dynamic Checking.

Answer:
Divide-by-zero errors, array index out of bounds, and errors caused by specific runtime values.

---

6. What are the benefits of Static Analysis?

Answer:
Early bug detection, improved security, enforcement of coding standards, and measuring code complexity.

---

7. What is the Hailstone Sequence?

Answer:
A sequence where if n is even, n = n/2, and if n is odd, n = 3n + 1, repeating until n reaches 1.

---

8. Why is Java considered a statically typed language?

Answer:
Because all variable types are checked and determined by the compiler before the program runs.

---

9. What is the difference between primitive types and object types in Java?

Answer:
Primitive types store actual values, while object types store references (memory addresses) to objects.

---

10. Why are Java primitive numeric types not true mathematical numbers?

Answer:
Because of integer division, integer overflow, and special floating-point values like NaN and Infinity.

---

11. Explain integer overflow in Java.

Answer:
When a calculation exceeds the maximum or minimum value of a type, the result wraps around silently without error.

---

12. What is Static Analysis able to detect in code?

Answer:
Type mismatches, uninitialized variables, dead code, resource leaks, and coding standard violations.

---

13. What is FindBugs?

Answer:
A static analysis tool that examines Java bytecode to detect bug patterns and classify defects by severity.

---

14. What are the main categories of bugs in FindBugs?

Answer:
Correctness, Bad Practice, Performance, Multithreaded Correctness, Security, and Malicious Code.

---

15. What is Checkstyle?

Answer:
A static analysis tool that focuses on enforcing Java coding style and formatting rules.

---

16. What is the difference between Arrays and Lists in Java?

Answer:
Arrays have fixed length, while Lists are variable-length and can grow dynamically.

---

17. Why can array overflow bugs be dangerous?

Answer:
Because they can cause runtime crashes and security vulnerabilities if not properly checked.

---

18. What is a mutable type?

Answer:
A type whose object values can be changed after creation.

---

19. Give examples of immutable types in Java.

Answer:
String, primitive types, wrapper classes, BigInteger, and BigDecimal.

---

20. What does mutating a value mean?

Answer:
Changing the internal contents of a mutable object, such as modifying a list or array.

---

21. What is an immutable reference?

Answer:
A reference declared with final that cannot be reassigned after initialization.

---

22. Can a final reference point to a mutable object?

Answer:
Yes, the reference cannot change, but the object it points to can still be mutated.

---

23. What is aliasing?

Answer:
Aliasing occurs when multiple variables reference the same mutable object.

---

24. Why is aliasing dangerous with mutable objects?

Answer:
Because changes through one reference affect all other references unexpectedly.

---

25. Why are immutable types safer than mutable types?

Answer:
They prevent unintended side effects, make programs easier to understand, and reduce bugs.

---

26. What problem occurs when modifying a list during iteration?

Answer:
A ConcurrentModificationException is thrown at runtime.

---

27. What is a snapshot diagram?

Answer:
A diagram that visualizes variables and the values or objects they reference at runtime.

---

28. Why is returning mutable objects risky?

Answer:
Because callers can modify the returned object and unintentionally change internal program state.

---

29. How can risks of mutation be reduced?

Answer:
By using immutable objects or returning copies of mutable objects.

---

30. Why is documenting assumptions important in programming?

Answer:
It improves readability, prevents misunderstandings, and helps maintain and modify code safely.